

Spectra

Documentation

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1. Getting started

1. Click on + to create a new project.
2. Double-click on canvas to create a new note.
3. Press space to play it back.
4. Press T to toggle partials.
5. Select a note. Press Up/Down keys to select following partials.
Press Shift+Up/Down to move it up/down in current tuning.
6. Select a partial with a mouse to observe the resulting behavior.
7. Press Shift+T to add a new tuning change.

This section is intended to reveal several key features of this program.

2. About Spectra

2.1 What is it?

Spectra is a spectral/microtonal sequencer rooted in thinking in partials rather than notes. It is a multi-functional environment that extends current possibilities of writing spectral and microtonal music and makes it much easier to write.

At first glance, Spectra is reminiscent of the look of Digital Audio Workstations, which is a deliberate design choice. It is, however, not focused on audio processing, but rather analysis and sound synthesis. Here are some unique aspects of this software:

- tuning, time segmentation and timbre are interconnected
- what you see is what you hear – audio is synthesized with additive synthesis in real-time
- possibility to change tuning at any point in a project
- hierarchy of partials does not exist, therefore default mode of vertical movement functions as search for the next frequency higher / lower partial

2.2 What Spectra can be

- **Composition tool** – either for full compositions or sketches, export to MusicXML
- **Microtonal lab** – quickly change, record, export microtonal ideas with custom/free tunings
- **Analysis tool** – analyze audio files and turn them to timbres/tunings
- **Format converter** – export/import timbres, tunings, .scl, .tun, grids, MIDI/OSC input/output
- **Educational tool** – OSC output enables control of multiple devices simultaneously.
- **Live performance tool** – Spectra can connect to controllers, process and output to software

3. Basics

The following section covers basic terminology.

The following is a brief overview of relevant acoustics. When singing a note, the resulting sound is in fact not a single pitch but a combination of many individual frequencies sounding simultaneously. The auditory system fuses individual simultaneously sounding frequencies into a timbre. The lowest frequency is a **fundamental**. The rest are **overtones**. For consistency, this documentation uses the term **partial** given its inclusion the fundamental tone. **Spectrum** (where the name for the software comes from) is these partials sounding together.

In spectral composition practice of the 70s/80s, one of the approaches involves analyzing a sound, extracting individual partials, and distributing them across instruments such that when played together, the result approximates the original sound. When placing notes in a notation program, a paper or a DAW, it becomes apparent that this is not quite straightforward. The standard Western notation assumes twelve equally spaced notes within an octave, but partials are specific multiples of the fundamental tone and do not align with these predefined twelve notes. Such intermediate pitches are referred to as **microtones**. Despite the prefix suggesting something small, the term refers to notes that do not align with a chromatic scale (12-EDO).

The following sections outline the core functionality of Spectra, followed by a comprehensive reference of all available features.

Upon creating your first project (clicking the **+** button, giving it a name and hitting Enter), you will be presented with an interface that resembles classic DAW midi-esque view with several panels.

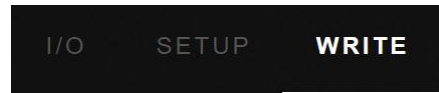
Top section is reserved for pages (I/O, Setup, Write) basic editing toggles, export button, volume control and help button.

I/O is reserved for configuring your real-time inputs and outputs and synchronization. This includes MIDI/OSC inputs/outputs, selecting whether you would just send fundamentals out to devices or all the partials at once, and so on.

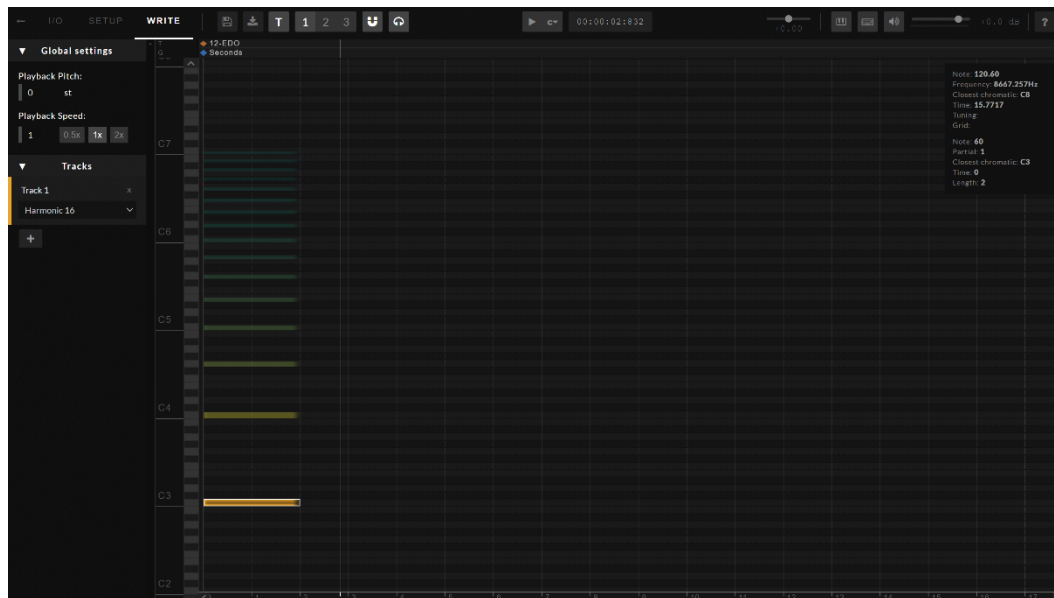
Setup is where you would create your timbres, tunings, grids and tools you can use in writing process. You have an absolute freedom in terms

of what kinds of tunings or timbres you want to create – repeating/non-repeating, custom pitches, from a sound analysis and more.

Write is the place where the actual music is written.

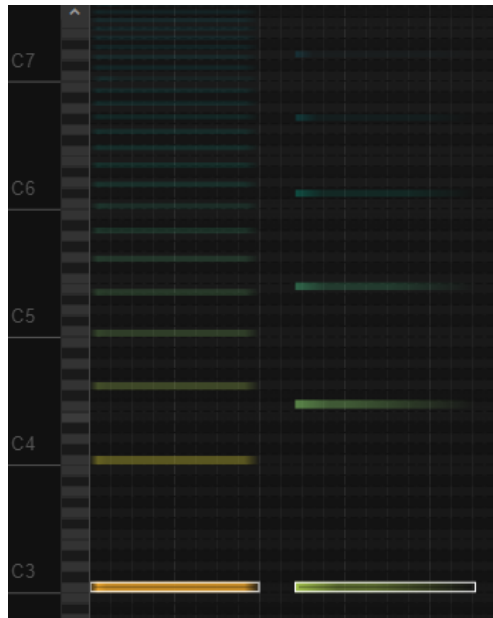


When opening the Write page, the layout is visually similar to established editors with piano roll, clear barlines, and placed notes that appear as rectangles. Editing features follow the same conventions, and Spectra contains all basic features such as copying, pasting, moving, dragging, resizing and duplicating.



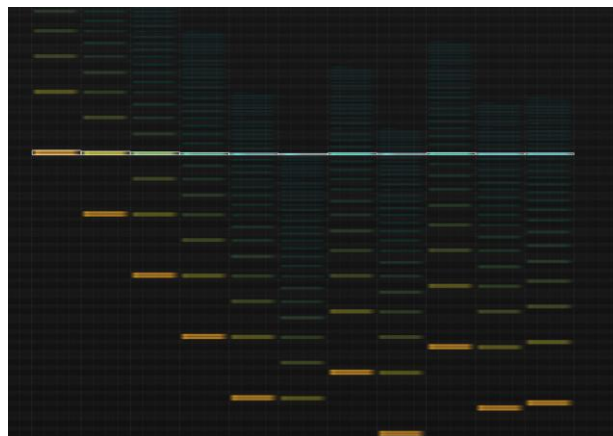
However, once you add a note by double-clicking, you'll notice that it is created with its own partials included.

Individual partials can be selected as well. This software does not hide the fact that relationships between partials are equally important as relationships between fundamentals. In notation, if an interval of C-F is played, where C is played by piano and F by vibraphone, notation doesn't care whether C-F is played by the same piano or two completely different instruments. Spectra does, and it shows the distinction explicitly.



Two notes played by two different instruments – sawtooth + bowl in this example

Upon creating a note and selecting it (visible by showing a dashed outline), press the Up arrow key. The selected partial moves slightly upwards while the fundamental changes. Shift+Up moves the note to the next step of the current tuning instead.



The first 11 stages of continuously pressing the Up arrow key

Pressing the Up/Down keys to find successive partials is referred to as **scrolling through partials**. It allows users to write music via partials while not thinking about the notes these partials came from. When scrolling through partials, you will also notice that the current partial is more

visible than others. This partial is called an **active partial** – it is purely a visual feature that has no effect on the resulting sound, but an aid for the user to determine which partial from the note's spectrum is the most important.



*Visual difference between an active partial
(marked with thick white outline and bright color)
and a highlighted/selected partial (dashed white outline)*

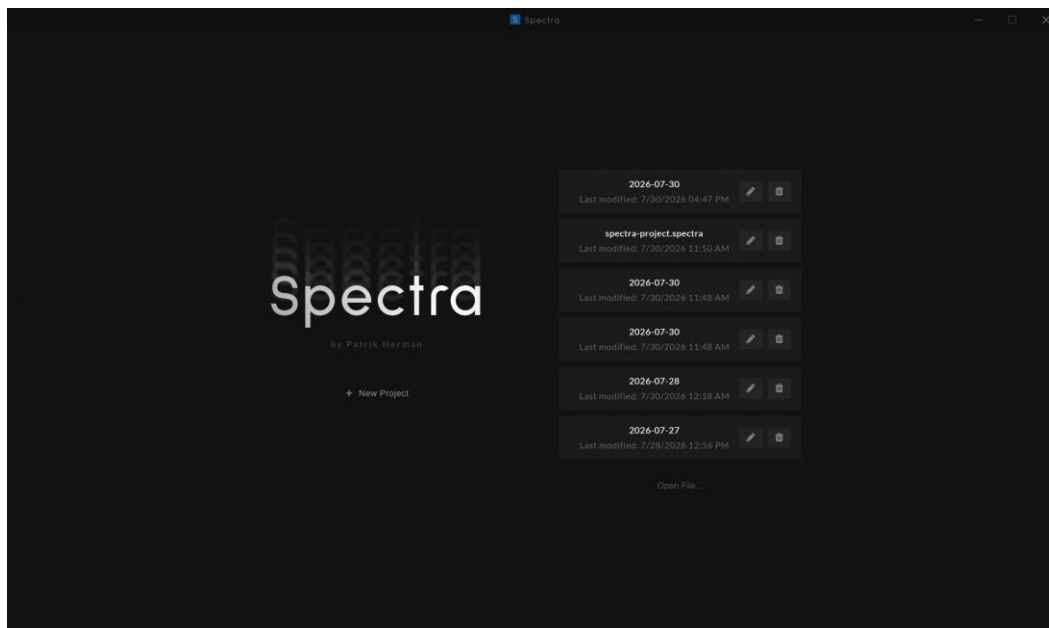
Summary of the terms introduced above:

- *scrolling through partials* means to find successive partials (slightly lower or higher) while not thinking about notes these partials came from
- *active partial* is the visual representation of the most important partial from the current spectrum
- a *spectrum* is a collection of together sounding partials
- *microtones* are in-between tones on a standard keyboard – in other words, notes outside 12-EDO

Press **T** to toggle inactive partials.

3.1 Project Management

Spectra stores all projects locally on the machine it runs on. Projects persist between sessions and require no account. To move a project to another machine, export it as a JSON file with the Save Project button and import it there.



Initial Spectra screen

To create a project, click the large **+** icon, enter your project name and press Enter key.

To rename a project, click the pencil icon.

To delete a project, click the trash bin icon.

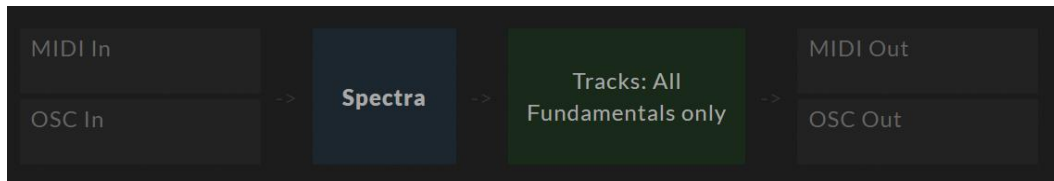
Importing projects:

- Spectra allows users to import both MIDI and JSON project files as new project.
- In order to import a project, click the **Open Project File...** button.

4. I/O Section

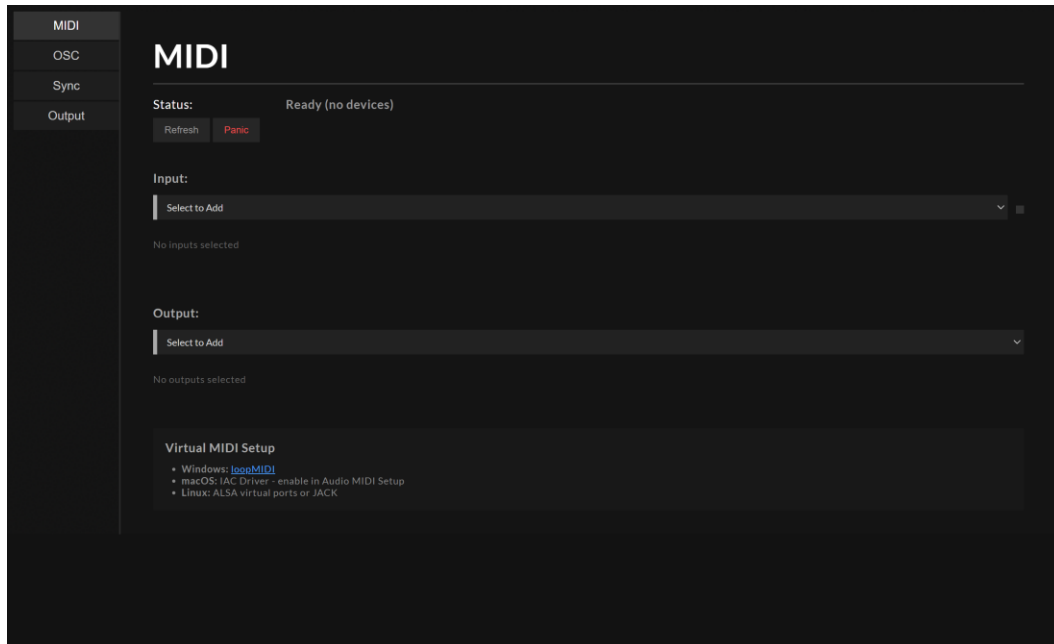
Spectra functions not only as a space for importing, writing, and exporting music, but also as middleware for connecting to external software and devices, and it uses MIDI/OSC to achieve this.

In I/O Section, the top section shows the currently active inputs and outputs:



4.1 MIDI I/O

The MIDI section contains several basic options to reload MIDI devices, a Panic button that stops all the played notes, and dropdowns to select MIDI inputs and outputs.



If you want to connect Spectra with your DAW or any software running on the same machine that can input and output MIDI, you need to use a virtual MIDI port. Each operating system requires a different configuration:

Operating system	Virtual MIDI
Windows	loopMIDI
macOS	IAC driver (built-in) in Audio MIDI Setup
Linux	ALSA virtual port, JACK

When an output device is chosen, upon starting playback, Spectra sends both MIDI noteOn and noteOff messages as well as specific SysEx 12-byte messages with the following format:

Microtonal Note (12 bytes)

```
F0 7D 01 [trk] [note] [vel] [pitL] [pitH] [sign] [partial] [amp] F7
```

Byte	Message
F0	SysEx message start
7D	Manufacturer ID
01	Message type (01 = note)
Track	Track index (0 - 127)
Note	Nearest MIDI note (0 - 127)
Velocity	Velocity (0 - 127, 0 = note off)
Pitch L/H	14-bit pitch deviation in decicents (0.1¢ resolution, ±1638.3¢ range)
Pitch Sign	0 = positive deviation, 1 = negative deviation
Partial num	Partial number (1 - 127)
Amplitude	Amplitude (0 - 127)
F7	SysEx message end

Panic / All Notes Off (12 bytes)

```
F0 7D 02 00 00 00 00 00 00 00 00 00 F7
```

Byte	Message
F0	SysEx message start
7D	Manufacturer ID
02	Panic message type
...	...
F7	SysEx message end

Transport Sync (12 bytes)

F0 7D 03 [timeL] [timeM] [timeH] [flags] 00 00 00 00 F7 (Start)
F0 7D 04 [timeL] [timeM] [timeH] [flags] 00 00 00 00 F7 (Stop)

Byte	Message
F0	SysEx message start
7D	Manufacturer ID
03/04	03 = Transport Start, 04 = Transport Stop
Time 0-3	28-bit time in milliseconds (max 74.5 hours)
Flags	Reserved flags for future use
...	
F7	SysEx message end

4.2 OSC I/O

Besides MIDI, Spectra also allows for OSC communication. In order to be able to send or receive OSC data, Spectra needs to pair with a device. The device is addressed directly over UDP: enter its IP address and port in the OSC settings, which default to 127.0.0.1 and port 9000. Spectra itself receives OSC on port 9002. Once a device answers, Spectra registers it and adds it to a list.

This device can be anything that is capable of sending/receiving UDP or Websocket data. Here are the OSC messages sent:

Devices are managed in a list: each entry has a name and a connection type, Scan Network looks for devices on the local network, and a pairing key can be regenerated at any time. Each spatial routing zone can be soloed or muted, and a zone is sent to the corresponding device in the OSC output list.

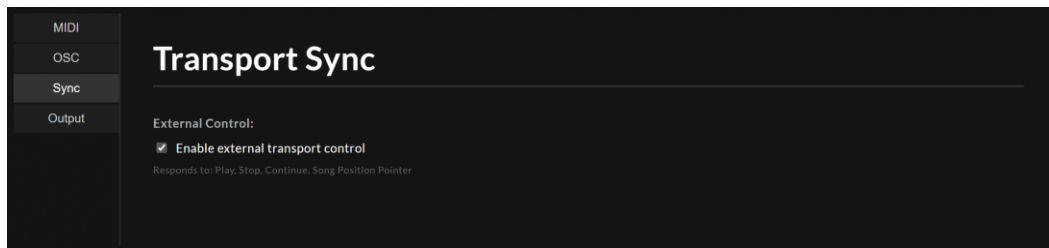
Incoming (to Spectra on port 9002):

```
/spectra/note/on pitch velocity [track]
/spectra/note/off pitch [track]
/spectra/playback playing(1/0) position
/spectra/note/on pitch velocity [track]
/spectra/note/off pitch [track]
/spectra/transport/start playing(1/0)
/spectra/transport/position position
/spectra/transport/loop enabled start end
/spectra/transport/speed multiplier
/spectra/transport/record enabled(1/0)
/spectra/write/note/create time pitch duration track [partial]
/spectra/write/note/delete time pitch track
/spectra/write/track/select id
/spectra/write/track/instrument id instrumentName
/spectra/write/tuning/add time tuningName [global]
/spectra/write/tuning/remove time
/spectra/write/tuning/change time tuningName [global]
/spectra/write/grid/add time gridName [global]
/spectra/write/grid/remove time
/spectra/write/grid/change time gridName [global]
```

Outgoing (from Spectra):

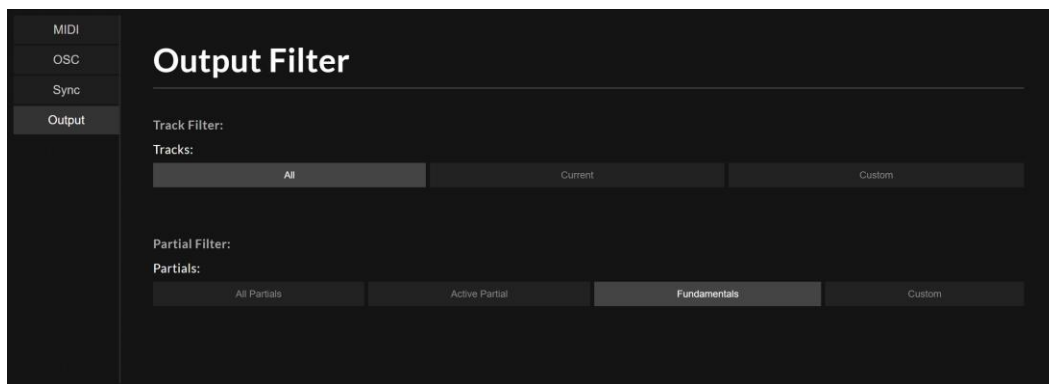
```
/spectra/note/on pitch velocity instrument partial
/spectra/note/off pitch instrument
/spectra/playback playing position
```

4.3 Transport Sync



Spectra is also capable of DAW/device synchronization. It sends MIDI Beat Clock at 24 PPQN (0xF8) together with Start (0xFA), Continue (0xFB) and Stop (0xFC), and a Song Position Pointer (0xF2) whenever the playhead moves. To use transport sync, enable MIDI sync in the DAW (the MIDI output selected in Spectra becomes the MIDI input in the DAW).

4.4 Output Filter



By default, the software sends fundamentals only to your external devices. However, if you wish to change this, you can send either all of the played partials (including fundamentals), active partials or a custom selection (e.g. only second and fourth partials of each played note).

As an example, if a composer wants to get partials of several retuned gongs, they would create a new timbre from extracted audio of a gong, place some notes down, set the Output Filter to All Partials and record the data being sent.

Spatial output routing is a feature unique to Spectra – because each timbre can have its own panning of its individual partials, you can consequently distribute individual partials in space. With spatial output

routing, you can decide what sections of the space get distributed to what outputs. For instance, if a timbre has an evenly distributed partials from lowest to highest and spatial output routing has three subsections, lowest partials would always be sent to the first output, middle partials in the second and highest partials into the third output.

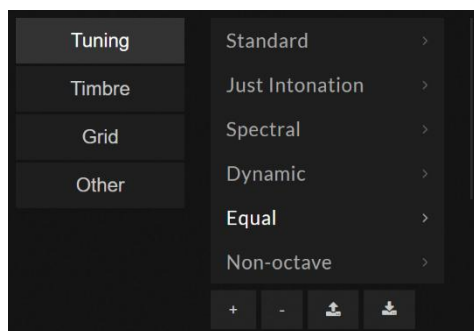
5. Setup

This page is one of the most essential parts of the entire software. Here you can create your own tunings, timbres, grids and edit them however you like. The Setup page contains four tabs: Tuning, Timbre, Grid, and Other.

Spectra comes with several built-in tunings and timbres to showcase its abilities. This includes some known EDO tunings as well as some timbre-based unconventional tunings.

The Other tab holds the settings that are not tied to a single tuning, timbre or grid. Defaults selects the tuning and the grid a new project starts with. Audio sets a latency compensation in milliseconds, which an Auto button can measure instead. MIDI Input sets the pitch center, the MIDI key the root of the tuning is mapped to, 60 by default. Partial Editing sets the partial limit, the highest number of partials drawn and sounded per note. Performance mode trades gradient rendering for speed. Two buttons delete stored data, either everything or everything except the settings.

5.1 Tuning



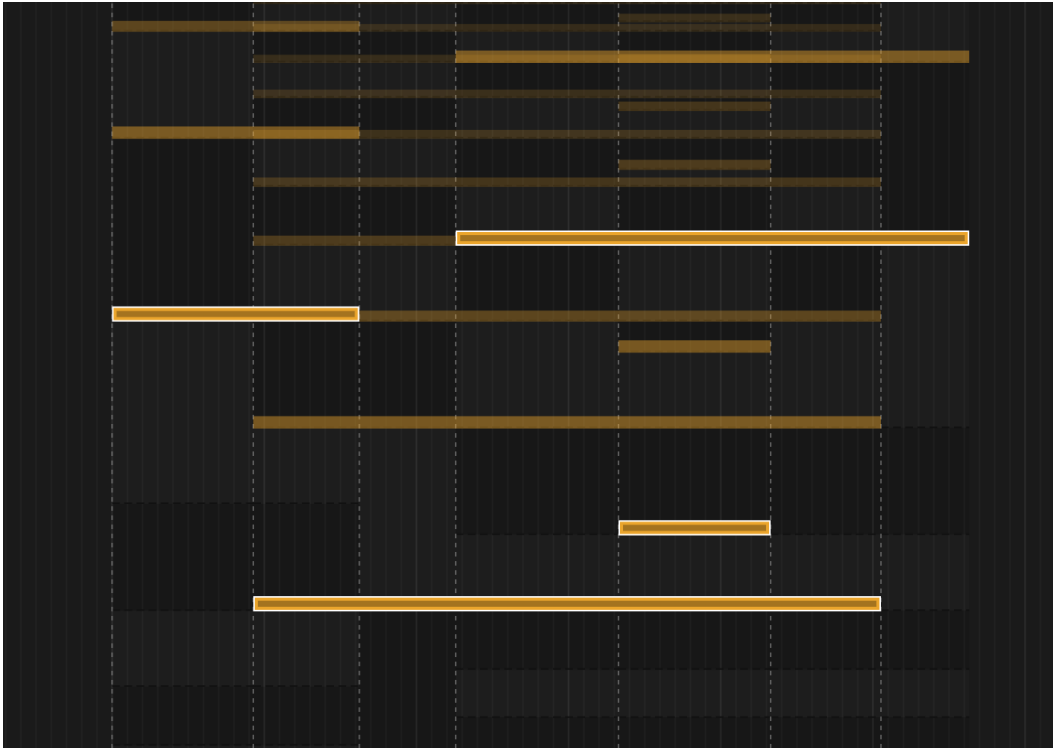
The software comes with several sections to generate tunings. The sections are Mathematical, Import, and Dynamic. Here are the following types of tuning generators:

1. **Equal Temperament** – equal division – choose the reference A, octave multiplier (default is 2x), and a number of steps/divisions. For instance, 31EDO would have octave multiplier 2 and number of steps 31. An approximation of Carlos Alpha would have an octave multiplier 1.5 and number of steps set to 9.
2. **Ratio Entry** – create a tuning from ratios entered one per line or separated by commas, with a reference frequency in Hz and the number of octaves the set is repeated over.
3. From Audio Analysis – Create a tuning extracted from analyzed partials of an audio file. A MIDI Offset field shifts the whole result up or down when it is mapped to the keyboard.

4. **File Import (.scl)** – Import a Scala tuning file as a tuning.
5. **Custom Frequencies** – If you want complete freedom, you can choose your own pitches for the tuning.
6. **Adaptive** – tuning that adjusts based on notes played.
7. **An equal division can be detuned step by step: the Custom Microtuning button opens an editor with one field per division, each holding a cent offset for that step, and Apply Microtuning writes them into the tuning.**

5.1.1 Adaptive tuning

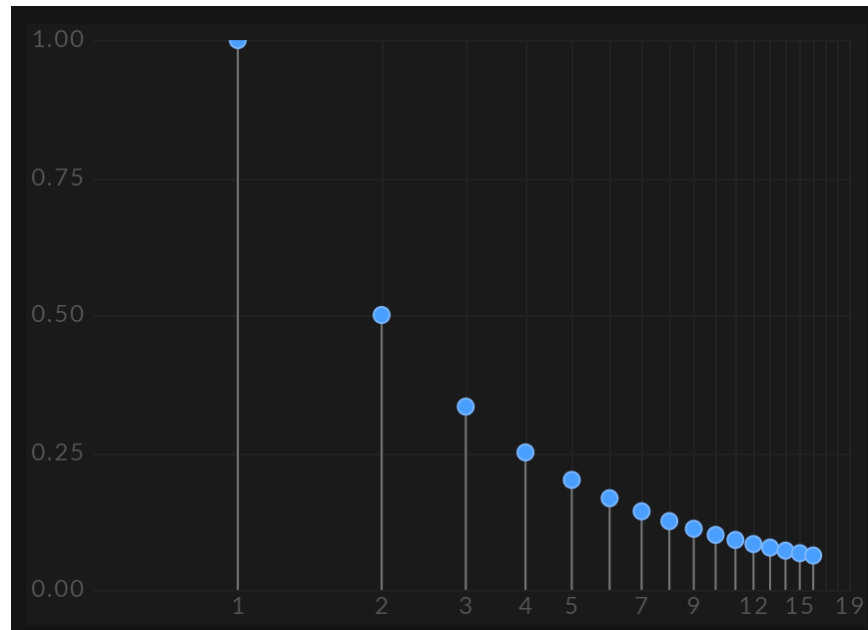
This tuning requires its own section. While conventional notation systems require users to first select a tuning and then place notes within that tuning, adaptive tuning reverses this relationship – the timbre of notes determines the momentary tuning. Adaptive tuning initially appears blank, but upon placing a note, Spectra creates tuning steps that are both coming from the timbre used as well as its inverted version – when using a simple sawtooth timbre, it is a simple Just intonation system, when using an inharmonic timbre, that is where unconventional tuning systems emerge. The Adaptive panel sets the lowest and highest frequency of the generated pitches and whether MIDI preview follows the adaptive tuning or snaps to the cursor.



Adaptive tuning in action – notes and their spectra determine momentary tuning

5.2 Timbre

A basic timbre is generated from the harmonic series, from odd harmonics only, or as a random spectrum with the inharmonic button. The graph below shows the individual harmonics and their amplitudes. You can drag individual points up/down and use Shift+Drag to move them around.



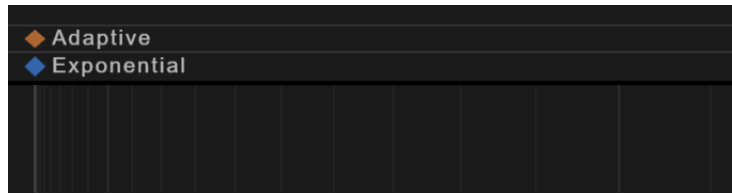
Timbre section also allows you to extract partials from an instrument, a sound, recording or any type of audio – by loading a wave file, selecting a portion of the audio and adjusting the sensitivity and threshold values, you can save these analysed partials as a custom timbre. Clicking the analysed frequencies or midi notes copies them to your clipboard.

The table of partials lists the exact multiples of the fundamental and allows editing them manually.

5.3 Grid

This section lets you create your own grids – by default, Spectra comes with a “seconds” grid that creates barlines at every second. Grids are not only a visual aid, but you can choose to snap or quantize notes to them. Here are the following types of grids:

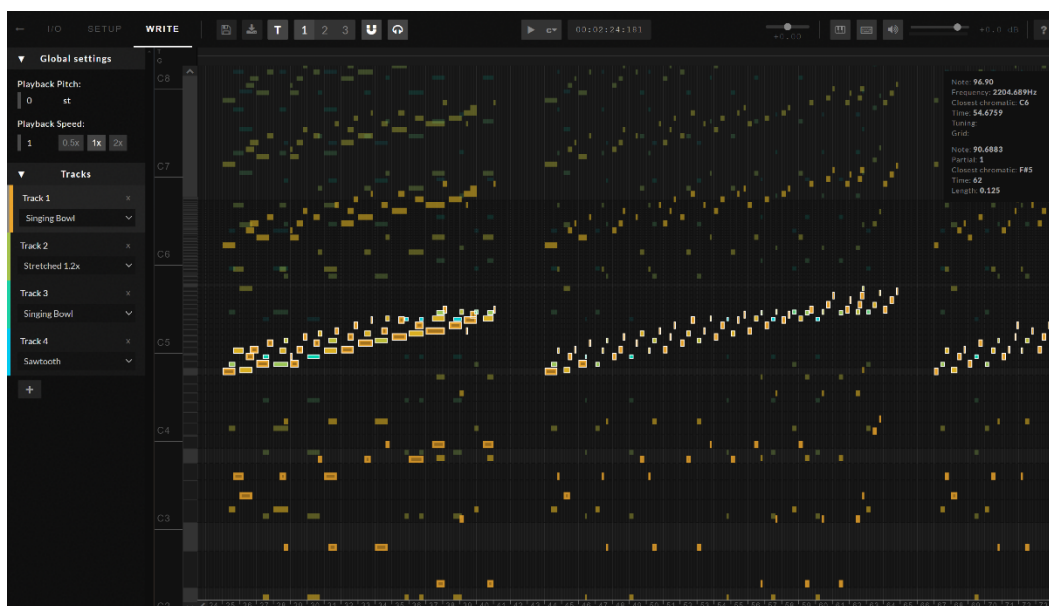
1. **Linear – Even spacing between gridlines. The spacing is entered either in milliseconds or as a BPM value, together with the number of subdivisions. Useful when making tuplets.**
2. **Exponential – Type of a grid where the spacing changes progressively with each bar.**



Exponential grid in action

6. Write

The Write page is where users are expected to spend the majority of their time. Despite the emphasis on non-standard tuning systems in earlier sections, the default view presents a standard grid with a 12-EDO system. This is a deliberate design choice intended to make the interface familiar to users of existing DAWs.

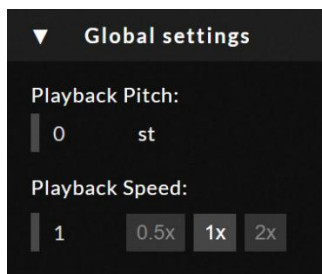


The **top panel** contains the most important toggles and switches. Each control is briefly described below. These controls are explained in detail in subsequent sections. These are:

- **Export** – letting the user download their project in various formats (audio, MIDI, MusicXML, JSON)
- **Save Project** – downloads the current project as a JSON file
- **T** – toggle inactive partials
- **1 / 2 / 3** – partial scrolling mode – Nearest Partial / Undertone Series / Hybrid
- **Headphones** – hearing changed / scrolled partials when editing
- **Magnet** – automatic note-snapping (**U** key)
- **Play** – Play / Stop playback
- **Playback mode** – Continue playback (the cursor continues from where it stopped) or Return back (the cursor returns to its original position)

- **Time** – show the exact time in format HH:MM:SS:MSS
- **Volume** – toggle audio on/off
- **Volume slider** – change output volume from -Inf to +6dB – Ctrl+Click resets the value
- **Brightness slider** – **adjusts which partials appear more prominent (Alt+Scroll, Ctrl+Click resets the value)**
- **Keyboard icon** – **toggle Computer Keyboard Mode**
Ctrl + M – toggle mute
- **?** – show keyboard shortcuts and other useful links and information

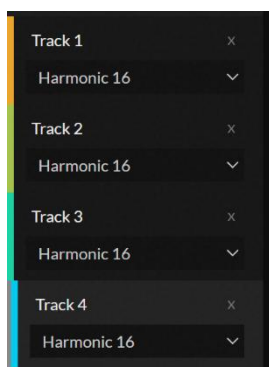
6.1 Global settings



Spectra offers a few global settings for the current project:

- **Playback pitch** – changes the tuning of playback. Defaults to 0.
- **Playback Speed** – changes playback speed. Defaults to 1x, can be set to 0.5x (half the speed) or 2x (twice the speed) or a custom value.

6.2 Tracks



Below Global settings in the left panel is a list of all of the tracks within the project. The plus button adds a new track, x deletes it. Each track is assigned an instrument which determines the spectra of its notes. Clicking on a track highlights notes contained in that track, the rest becomes grey. Shift+Click and Ctrl+Click allows you to select multiple tracks at the same time.

Double-click the track's name to change it.

By default, Spectra uses OKLAB color space to choose its track colors for new tracks, and you can choose a custom color by clicking on the coloured strip on the left side of the track.

When changing track's instrument, Spectra attempts to find the closest match for the active partial to still preserve your musical content.

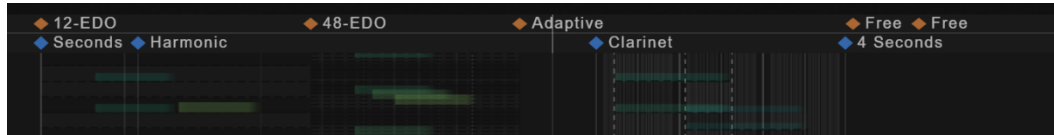
Tracks are independent by default – instruments are not linked, while tuning and grid changes apply to every track unless a single event is limited to one. As an example, your Track 1 can be in 24EDO while Track 2 is in 31EDO.

There are technically no constraints on the track limit, however, MIDI SysEx data supports sending notes from the first 128 tracks.

6.3 Undo/Redo

Majority of functions are linked to an event history (of a size of 100 events) such that you can undo/redo using Ctrl+Z to undo and Ctrl+Y/Ctrl+Shift+Z to redo.

6.4 Tuning / Grid events



Two lanes sit above the main canvas: tuning changes and grid changes.

Tuning / Grid events are unique to Spectra.

One of the main philosophies of Spectra is to allow users to be unrestrained by basic tuning and grid conventions that DAWs and notation programs provide. Even a basic system like a five-line staff assumes the composer wants to write in a diatonic system. This software solves this not only with free tunings and grids, but tuning / grid events that let user change tunings and grids at any point in the project.

- To add a tuning / grid event, double-click on the corresponding lane and choose your tuning or a grid.
- Events are draggable with mouse by default. Shift+Drag snaps them to the current grid.

- A tuning or grid event applies to every track: the Global (applies to all tracks) box in its panel is ticked by default. Untick it to limit the event to the track it sits on.

6.5 Main Canvas, Note Editing

The main canvas not only contains all the basic note editing shortcuts but also comes with a few useful shortcuts specific to this tool. The main difference between Spectra and standard MIDI note editing tools is how successive notes are selected, which is determined by the current partial scrolling mode. The following shortcuts are designed for efficient note editing. Important shortcuts or shortcuts unique to this software are marked in **bold**.

Creating, Deleting notes

- Double-click – create a note
- Double-click + drag – create a note with custom length
- Del / Backspace – delete selected note(s)
- D – duplicate selected note(s)
- **Ctrl + Drag – duplicate selected note(s) while dragging**

Note selection

- Click – select a note or a partial
- Shift + Click – select multiple notes or partials
- Drag empty area – rectangle selection
- Ctrl + A – select all notes in the selected tracks

Note resizing

- Drag left / right side of note – resize selected note(s)
- Shift + Left/Right – resize selected note(s) (when notes selected); skip to timeline event (when no notes selected)

Note dragging

- Drag – move the note forwards / backwards in time or move it to the closest step in tuning
- **Shift + Drag – snap selected note(s) while moving**

Keyboard

- Up / Down – move to the next partial depending on partial scrolling mode
- Shift + Up / Down – move to the next step in the current tuning
- Ctrl + Shift + Up / Down – move the note(s) up / down an octave
- Ctrl + Up / Down – select the nearest higher / lower partial of the selected note(s)
- Alt + Up / Down – move the selected note(s) by one cent
- Ctrl + Alt + Up / Down – move the selected note(s) by ten cents

Playhead

- Left / Right – move by one grid step (when no notes selected); move selected note(s) in time (when notes selected)
- Shift + Left / Right – skip to next/previous timeline event (when no notes selected)

Quantization

- Q – quantize note(s) to the current grid
- **Shift + Q – quantize note(s) to the current tuning**
- Shift + T – add a tuning change at the playhead position**
- Shift + G – add a grid change at the playhead position**

Copy, Paste

- Ctrl + C – copy selected note(s)
- Ctrl + V – paste selected note(s)
- Ctrl + X – cut selected note(s)
- Ctrl + S – save project
- Ctrl + M – toggle mute
- Ctrl + 1 / 2 / 3 – switch to I/O / Setup / Write tab
- Ctrl + Shift + K – panic: stops every sounding note and sends All Notes Off to the MIDI output
- Space – play / pause playback

Visibility

Since each note displays its full spectrum, visual density can increase significantly. Spectra handles this with several shortcuts:

- **T – toggle visibility of inactive partials. While they are hidden, only the active partial of a note can be selected, and a rectangle selection picks up nothing else**

Partial Locking

Another feature that is unique to Spectra is the ability to lock partials in place. When a partial is locked, when selecting a different partial from the same tuning, Spectra attempts to find a note where its spectrum contains this frequency of the locked partial while at the same time contains another partial that is slightly above or below the current active partial. In simple terms, if you are writing two-part harmony with partials but only with a single continuous line of notes, this tool allows you to do this quite easily.

- **L – lock the selected partial**

Magnet mode

Magnet mode snaps notes by their edges.

- U – toggle magnet mode on / off
- **M – toggle Computer Keyboard Mode (play notes with letter keys)**
- **, and . – shift the Computer Keyboard Mode octave down / up**

Preview while editing

- **P – play the fundamental tone of selected note(s)**
- **O – play the partial of the selected note(s)**
- **H – toggle previewing partials while note editing**
- S (hold) – Spotlight mode: ducks master volume to -30dB and plays the hovered partial in isolation. Releases when S is released.**

Partial selection mode

When highlighting partials and pressing Up / Down keys, Spectra finds closest partials depending on a selected mode. These are:

- **1, Nearest Partial – finds the next partial whose fundamental comes from the current tuning**
- **2, Undertone Series – scrolls through the utonal series while the active partial stays in place**
- **3, Hybrid – combination of both modes**

Brightness Offset

Alt + Scroll adjusts the brightness offset of partials, changing which partials are more prominent. Ctrl + Click on the brightness control resets it to the default value.

MIDI Partial Mode

Spectra includes a MIDI Partial Mode toggle that, when enabled, plays ordered partials from the current timbre instead of tuning steps when receiving MIDI input. This is useful for exploring the spectrum of an instrument directly from a MIDI controller.

An information panel next to the canvas follows the cursor and reads out the pitch as a fractional MIDI number, its frequency in Hz, the closest chromatic note name, the time in seconds, and the tuning and grid in force at that point. When the cursor rests on a partial, a second block adds that partial number, its own closest chromatic note, and its start and length.

Mouse navigation

- Scroll – scroll vertically (pitch)
- Shift + Scroll – scroll horizontally (time)
- Ctrl + Scroll – zoom in / out in time
- **Ctrl + Shift + Scroll – zoom vertically (pitch)**
- **Ctrl + Alt + Scroll – zoom in / out both in time and pitch**

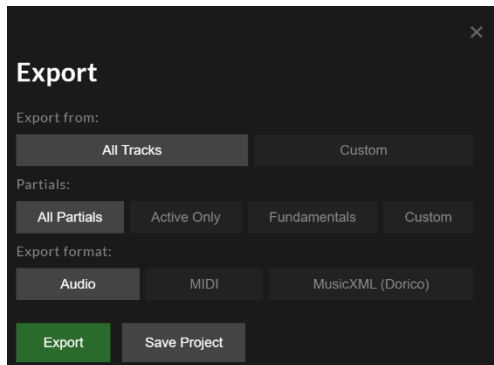
Spectra comes with basic scroll functions. You can also zoom in / out on time area (numbered section at the very bottom of the canvas) as well as keyboard area. Double-clicking resets the zoom level. The same movements are available as buttons next to those two areas, four for zooming in and out in either axis and four for panning.

6.6 Timeline

At the bottom of the main canvas is the timeline, showing time in seconds. The playhead (vertical line) indicates the current playback position.



7. Exporting



To open the export window, either press Ctrl + Shift + E or click the export button right next to Write button in the top menu.

Export window contains several options:

Export from: 1. All Tracks – exported data contains data from all the

tracks 2. Custom – choose which tracks get exported

Partials:

From the selected tracks, select which partials get exported.

- All Partials – all partials get exported as notes.
- Active Only – only visually highlighted partials get exported.
- Fundamentals – Only fundamental tones get exported. For example, if your external synth contains your track's spectrum with a complex sound design, exporting fundamentals gets you to the same result, but with sound design added.
- Custom – choose which partial numbers get exported.

Export format:

- **Audio** – exports as .wav file.
- **MIDI** – exports as a .mid file containing the tracks
 - o Microtonal pitches are approximated to the nearest semitone.
 - o Check Pitch bend checkbox to preserve pitch bending in MIDI or leave MIDI quantized to chromatic scale.
 - o Useful for quick imports to other DAWs and devices capable of reading MIDI.

- **MusicXML** – exports as a .musicxml file to be imported by notation software.
 - o Supported for Dorico
 - o When choosing this option, you have an option to choose pitch quantisation – either Off (exact pitch content is preserved) or From Tuning (pitches are quantised to the steps of the tuning in use).

Project export:

- **Save project** button downloads the .spectra file of the entire project.
- This file includes all the notes, timbres, tunings, grids and settings and can be re-imported into Spectra.
- This is useful to share a project between devices.
- Tunings, timbres and grids are exported separately from the Setup page, each with its own export button. A tuning can be written as JSON, as a Scala scale (.scl), as an AnaMark tuning (.tun) or as a Scala keyboard mapping (.kbm); timbres and grids are written as JSON. The .tun and .kbm formats need a center MIDI note, which Spectra asks for before writing the file: the closest tuning note is mapped to that key and the rest are mapped outward from it.

8. Other

Supported platforms

Spectra runs as a standalone desktop application built with Electron. It requires no browser, works without a network connection and communicates OSC natively over UDP.

Offline usage

The application works fully offline. OSC communication reaches devices on the local network and needs no internet access.

9. FAQ

Where are my projects stored?

- Projects are stored locally by the application and persist until they are deleted manually.

Can I use Spectra offline?

- Yes. The application works fully offline.

10. Glossary of terms

- **Active partial** - visual representation of the most important partial from the current spectrum
- **Fundamental** - lowest frequency of a given spectrum.
- **Harmonic** - integer multiplier of the fundamental.
- **Overtone** - frequencies above the fundamental.
- **Partial** - individual frequency of a sound.
- **Spectrum** - collection of partials sounding together.
- **Scrolling through partials** - choosing next/previous partials. Pressing the Up / Down keys on selected notes makes Spectra choose the next relevant partial. In mode 1, it chooses next partial where the fundamental comes from the current tuning. In mode 2, it chooses next partial in utonal/subharmonic series. In mode 3, it chooses next partial from both arrays. When another partial from the same spectrum is locked, it chooses next higher or lower partial such that there is a partial at the frequency of the locked partial while finding the next partial in a row.
- **EDO/TET** - Equal Division of an Octave / ...-tone equal temperament.
- **Cent** - 1200th of an octave or 100th of a semitone.
- **Beating** - Interference pattern / amplitude cancellation with an audible pulse that is caused by a combination of two sinewaves with similar frequencies.
- **Reinforcement** - volume sum of two simultaneously sounding sinewaves caused by small frequency difference.